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Structural Valve Deterioration After Self-Expanding Transcatheter or Surgical Aortic Valve Implantation in Patients at Intermediate or High Risk

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IMPORTANCE The frequency and clinical importance of structural valve deterioration (SVD) in patients undergoing self-expanding transcatheter aortic valve implantation (TAVI) or surgery is poorly understood.

OBJECTIVE To evaluate the 5-year incidence, clinical outcomes, and predictors of hemodynamic SVD in patients undergoing self-expanding TAVI or surgery.

DESIGN, SETTING, AND PARTICIPANTS This post hoc analysis pooled data from the CoreValve US High Risk Pivotal (n = 615) and SURTAVI (n = 1484) randomized clinical trials (RCTs); it was supplemented by the CoreValve Extreme Risk Pivotal trial (n = 485) and CoreValve Continued Access Study (n = 2178). Patients with severe aortic valve stenosis deemed to be at intermediate or increased risk of 30-day surgical mortality were included. Data were collected from December 2010 to June 2016, and data were analyzed from December 2021 to October 2022.

INTERVENTIONS Patients were randomized to self-expanding TAVI or surgery in the RCTs or underwent self-expanding TAVI for clinical indications in the nonrandomized studies.

MAIN OUTCOMES AND MEASURES The primary end point was the incidence of SVD through 5 years (from the RCTs). Factors associated with SVD and its association with clinical outcomes were evaluated for the pooled RCT and non-RCT population. SVD was defined as (1) an increase in mean gradient of 10 mm Hg or greater from discharge or at 30 days to last echocardiography with a final mean gradient of 20 mm Hg or greater or (2) new-onset moderate or severe intraprosthetic aortic regurgitation or an increase of 1 grade or more.

RESULTS Of 4762 included patients, 2605 (54.7%) were male, and the mean (SD) age was 82.1 (7.4) years. A total of 2099 RCT patients, including 1128 who received TAVI and 971 who received surgery, and 2663 non-RCT patients who received TAVI were included. The cumulative incidence of SVD treating death as a competing risk was lower in patients undergoing TAVI than surgery (TAVI, 2.20%; surgery, 4.38%; hazard ratio [HR], 0.46; 95% CI, 0.27-0.78; P = .004). This lower risk was most pronounced in patients with smaller annuli (23 mm diameter or smaller; TAVI, 1.32%; surgery, 5.84%; HR, 0.21; 95% CI, 0.06-0.73; P = .02). SVD was associated with increased 5-year all-cause mortality (HR, 2.03; 95% CI, 1.46-2.82; P < .001), cardiovascular mortality (HR, 1.86; 95% CI, 1.20-2.90; P = .006), and valve disease or worsening heart failure hospitalizations (HR, 2.17; 95% CI, 1.23-3.84; P = .008). Predictors of SVD were developed from multivariate analysis.

CONCLUSIONS AND RELEVANCE This study found a lower rate of SVD in patients undergoing self-expanding TAVI vs surgery at 5 years. Doppler echocardiography was a valuable tool to detect SVD, which was associated with worse clinical outcomes.

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Transcatheter aortic valve implantation (TAVI) has been established as an alternative to surgery in patients of all risk levels with symptomatic severe aortic stenosis (AS).[1-8] Current guidelines support a heart team discussion of the relative risks and benefits of surgery and TAVI in patients between ages 65 and 80 years.[9,10] Lifetime management after aortic valve replacement is an important part of this discussion, particularly in younger patients,[11] with bioprosthetic valve durability being a central theme to avoid recurrent symptoms or the need for a reintervention.[12]

Standardized definitions of bioprosthetic valve dysfunction have been proposed and categorized into structural valve deterioration (SVD) (ie, permanent valve changes leading to AS or intraprosthetic aortic regurgitation [AR]), nonstructural valve dysfunction (ie, paravalvular regurgitation or prosthesis-patient mismatch), thrombosis, and endocarditis.[13,14] Although SVD is a key component of bioprosthetic valve durability, scarce data exist on the incidence and factors associated with SVD after TAVI and surgery from large-scale multicenter randomized clinical trials (RCTs). One small randomized study showed a lower incidence of SVD in patients treated with a self-expanding supra-annular transcatheter bioprosthesis compared with surgery at 8 years.[15] A meta-analysis of prior randomized studies found lower rates of SVD with a supra-annular CoreValve bioprosthesis (Medtronic) compared with either surgery or a balloon-expandable intra-annular transcatheter bioprosthesis.[16]

This post hoc analysis evaluated the 5-year incidence and predictors of SVD as well as the association between SVD and clinical outcomes in patients undergoing self-expanding supra-annular TAVI or surgery from the CoreValve US High Risk Pivotal and SURTAVI trials.

Key Points

Question What are the 5-year incidence, outcomes, and predictors of structural valve deterioration (SVD) after supra-annular, self-expanding transcatheter aortic valve implantation (TAVI), or surgery from large-scale randomized clinical trials?

Findings In this analysis of pooled data from 2 randomized clinical trials, among 2099 randomized patients with severe aortic stenosis, the 5-year rate of SVD was 4.38% in patients receiving surgery and 2.20% in patients receiving TAVI. The Doppler-derived SVD imparted a 2-fold risk for all-cause mortality and hospitalization for valve disease or worsening heart failure.

Meaning In this study, Doppler echocardiography was a valuable tool to detect SVD and was associated with worse clinical outcomes.

Methods

Pooled Trial Design

Clinical and echocardiographic outcomes from the CoreValve US High Risk Pivotal (n = 615)[4] and SURTAVI (n = 1484)[6] RCTs were used to compare the rates of SVD at 5 years in patients undergoing CoreValve/Evolut R TAVI or surgery. To identify late clinical outcomes and predictors associated with SVD, data from these trials were supplemented with 5-year outcomes from the CoreValve US Extreme Risk Pivotal single-arm trial[2,17] (n = 485) and the single-arm CoreValve Continued Access Study (CAS; n = 2178). The primary outcomes of the RCT and non-RCT studies[2,4,6] and the 5-year outcomes of the RCTs[17-19] have been reported in detail elsewhere.

Study Conduct

Participating sites, investigators, and clinical protocols are found in the primary publications.[2,4,6] The trials were designed by the trial sponsor and overseen by the respective steering committees. All protocols were approved by the respective institutional review board or ethics committee at each site, and all patients provided written informed consent. The sponsor funded all trial-related activities and participated in site selection, data collection and monitoring, and statistical analysis. These studies were conducted in compliance with the International Conference on Harmonisation and the Declaration of Helsinki. The principal investigators and steering committees monitored all aspects of trial conduct.

Patient assessments were performed at baseline, discharge, 30 days, 6, 12, and 18 months, and annually through 5 years postprocedure. Clinical events were adjudicated by independent clinical events committees.[2,4,6] A single independent Echocardiographic Core Laboratory (Mayo Clinic, Rochester, Minnesota) evaluated protocol-mandated echocardiograms at baseline, discharge, 30 days, 6 months, and annually through 5 years. All available Core Laboratory-assessed echocardiograms were used in the analysis. When Core Laboratory assessment was not available, clinical site-reported echocardiographic readings were used. Core Laboratory echocardiograms were not collected at years 3 and 4 for the RCTs and at years 3, 4 and 5 for the Extreme Risk Pivotal trial. CoreValve CAS only had available site-reported echocardiographic readings. Echocardiograms were not collected at 30 days for the SURTAVI RCT. In case of a reintervention, the last echocardiogram before the reintervention was used.

Study End Points

The primary end point was the incidence of moderate or greater hemodynamic SVD through 5 years. Moderate SVD was defined as (1) hemodynamic valve deterioration (HVD) showing an increase in mean aortic gradient of 10 mm Hg or greater from discharge or 30-day echocardiography to last available echocardiography with a final mean gradient of 20 mm Hg or greater or (2) new occurrence or increase of 1 grade or more of intraprosthetic AR resulting in moderate or severe AR. Severe SVD was defined as (1) HVD showing an increase in mean gradient of 20 mm Hg or greater from discharge or 30-day echocardiography to last available echocardiography with a final mean gradient of 30 mm Hg or greater or (2) new occurrence or increase of 2 grades or more of intraprosthetic AR resulting in severe AR.[13,14] All potential SVD cases were verified by an algorithm established and validated by a group of 5 experts (S.J.Y., K.J.G., J.K.O., S.I., and M.J.R.). Additional criteria for SVD per Valve Academic Research Consortium (VARC-3)[14] and for HVD

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Table. Baseline Clinical Characteristics

Characteristic	Patients, No. (%)[a]		
	Surgery RCT (n = 971)	TAVI RCT (n = 1128)	TAVI non-RCT (n = 2663)[b]
Age, mean (SD), y	80.6 (6.3)	80.9 (6.5)	83.1 (8.0)[c]
Sex			
Female	444 (45.7)	496 (44.0)	1217 (45.7)
Male	527 (54.3)	632 (56.0)	1446 (54.3)
Body surface area, mean (SD), m^2	1.9 (0.2)	1.9 (0.2)	1.9 (0.3)[c]
STS-PROM, mean (SD)[d]	5.3 (2.5)	5.2 (2.4)	8.7 (4.6)[c]
NYHA HF class III/IV	639 (65.8)	757 (67.1)	2288 (85.9)[c]
Prior percutaneous coronary intervention	253 (26.1)	280 (24.8)	1052 (39.5)[c]
Prior coronary artery bypass surgery	213 (21.9)	229 (20.3)	973 (36.5)[c]
Hypertension	889 (91.6)	1056 (93.6)	2458 (92.3)
Creatinine >2.0 mg/dL	24 (2.5)	24 (2.1)	121 (4.5)[c]
Prior atrial fibrillation/flutter	305 (31.4)	348 (30.9)	1132 (42.6)[c]
Baseline anticoagulation therapy	236 (24.3)	236 (20.9)	558 (21.0)

Abbreviations: HF, heart failure; NYHA, New York Heart Association; RCT, randomized clinical trial; STS-PROM, Society of Thoracic Surgeons Predicted Risk of Mortality; TAVI, transcatheter aortic valve implantation.

SI conversion factor: To convert creatinine to μmol/L, multiply by 88.4.

[a] There were no significant differences between the surgery and TAVI RCT populations.

[b] The non-RCT TAVI cohort comprises the pooled CoreValve US Extreme Risk and the CoreValve CAS populations.

[c] $P < .001$ vs TAVI RCT.

[d] STS-PROM provides an estimate of the risk of death at 30 days among patients undergoing surgical aortic valve replacement based on several demographic and procedural variables.

due to changes in gradient alone[20,21] are found in eMethods 1 in the Supplement.

Statistical Analysis

Categorical variables are reported as counts and frequencies and compared using the χ^2 or Fisher exact test, where appropriate. Continuous variables are presented as means and SDs and compared using the *t* test. For ordinal data, the Cochran-Mantel-Haenszel test was used. The cumulative incidence rate of SVD at 5 years was calculated for the surgery and TAVI RCT populations using interval censoring analysis and treating death as a competing risk; treatment differences were summarized with a Fine-Gray proportional subdistribution *P* value (eMethods 2 in the Supplement).[22]

The association between SVD and clinical outcomes and predictors of SVD analyses were performed for the pooled surgery RCT and all TAVI (RCT and non-RCT) populations and separately for the surgery RCT and all TAVI cohorts. Univariate Cox proportional hazard models were performed with SVD as a time-dependent covariate to calculate the association of SVD with all-cause mortality, cardiovascular mortality, hospitalization for aortic valve disease or worsening heart failure, and the composite of mortality or hospitalization.

Univariate and multivariate analyses were performed to identify baseline clinical predictors of SVD using Fine-Gray proportional subdistribution hazards models for interval censored data with death as a competing risk. The final multivariate model was obtained using backward elimination with stay criteria of $P = .10$. No adjustments were made for multiple comparisons. Results were considered statistically significant at $P < .05$, and all *P* values were 2-sided. All statistical analyses were performed using the SAS software version 9.4 (SAS Institute) and R version 4.0.3 (The R Foundation).

Results

Of 4762 included patients, 2605 (54.7%) were male, and the mean (SD) age was 82.1 (7.4) years. The comparison analysis of SVD rates between TAVI and surgery populations included 971 patients randomized to surgery and 1128 patients randomized to TAVI. The analysis cohort for the predictors and clinical outcomes associated with SVD included the randomized patients supplemented with an additional 2663 patients who received TAVI that were treated in the non-RCT studies (eFigure 1 in the Supplement). Baseline characteristics of these cohorts are found in the **Table**. There were no significant differences between the RCT cohorts, but non-RCT patients who received TAVI had more baseline comorbidities compared with the TAVI RCT population (Table). The type and size of the surgical valves used in this pooled analysis are reported in eTable 1 in the Supplement. In the RCTs, the CoreValve bioprosthesis was used in 998 patients (88.5%) and the Evolut R bioprosthesis was used in 130 patients (11.5%). In the non-RCT studies, the CoreValve bioprosthesis was implanted in all patients. The median (range) follow-up time from index procedure to last available echocardiogram was 48.0 (1.8-98.4) months for the RCT surgery arm, 49.0 (4.6-97.9) months for the RCT TAVI arm, and 33.8 (0.2-68.7) months for non-RCT TAVI arm.

Echocardiographic Findings

Through 5 years, transvalvular mean gradients were significantly lower and effective orifice areas (EOA) were significantly larger for patients receiving TAVI compared with surgery at all time points postprocedure (**Figure 1**). The Doppler velocity index was significantly higher for patients receiving

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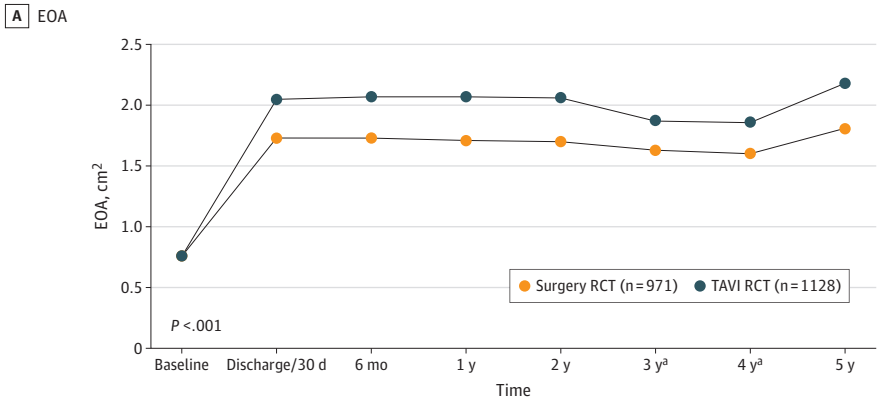
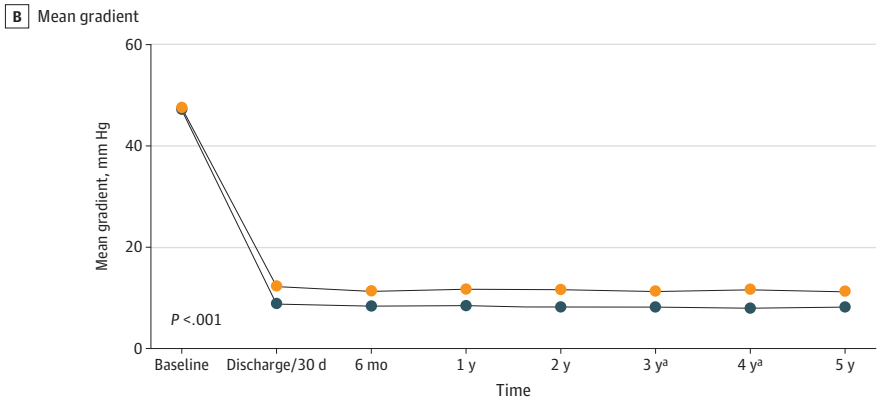
Figure 1. Hemodynamics in Patients Randomized to Surgery or Transcatheter Aortic Valve Implantation (TAVI)



No. at risk	Baseline	Discharge/30 d	6 mo	1 y	2 y	3 y[a]	4 y[a]	5 y
Surgery EOA	919	705	821	752	649	558	456	266
TAVI EOA	1061	951	989	930	788	702	579	434



No. at risk	Baseline	Discharge/30 d	6 mo	1 y	2 y	3 y[a]	4 y[a]	5 y
Surgery gradient	966	872	898	829	725	620	512	405
TAVI gradient	1122	1026	1071	1007	882	769	644	499

Effective orifice area (EOA) and mean gradient hemodynamic trends through 5 years. Patients in the TAVI group had significantly larger EOA and significantly lower mean gradient than patients in the surgery group at all time points after the procedure. RCT indicates randomized clinical trial.

[a] Change from Core Laboratory to site-reported echocardiographic readings.

TAVI compared with surgery immediately after the procedure (eTable 2 in the Supplement). The frequency of severe prosthesis-patient mismatch per VARC-3 was significantly lower in patients receiving TAVI than surgery after the procedure (eTable 2 in the Supplement).

SVD

SVD was identified in 95 of 4762 patients through 5 years (RCT, 37 receiving surgery and 21 receiving TAVI; non-RCT, 37 receiving TAVI). Echocardiographic findings of patients who developed SVD were similar among patients receiving surgery and TAVI (eTable 3 in the Supplement). The cumulative incidence rate of SVD treating death as a competing risk was significantly lower following TAVI than surgery (surgery, 4.38%; TAVI, 2.20%; hazard ratio [HR], 0.46; 95% CI, 0.27-0.78; pooled $P = .004$; P adjusted by study = .005) in the randomized patients (**Figure 2**A). This relative reduction in SVD was more pronounced in patients with a smaller annuli (computed tomography perimeter-derived diameter of 23 mm or less; surgery, 5.84%; TAVI, 1.32%; HR, 0.21; 95% CI, 0.06-0.73; $P = .02$) than in patients with a larger annuli (computed tomography perimeter-derived diameter greater than 23 mm; surgery, 3.99%; TAVI, 2.50%; HR, 0.57; 95% CI, 0.32-1.04; $P = .07$) (Figure 2B and C). RCT patients receiving TAVI had a numeric reduction compared with those receiving surgery in patients with both moderate and severe SVD (Figure 2D). The 5-year incidence rate of severe SVD was similar after surgery and TAVI (surgery, 0.74%; TAVI, <0.01%; HR, 0.40; 95% CI, 0.10-1.59; $P = .19$). The low numbers of severe SVD events based on severity at last available echocardiogram (RCT, 6 receiving surgery and 3 receiving TAVI) prevented to detect a statistically relevant difference. Rates when alternative SVD definitions per VARC-3 and due to changes in gradient alone were used are reported in eFigure 2 in the Supplement.

Clinical Outcomes With SVD

Patients who developed SVD had a significant increase in 5-year all-cause mortality (HR, 2.03; 95% CI, 1.46-2.82; $P < .001$), cardiovascular mortality (HR, 1.86; 95% CI, 1.20-2.90; $P = .006$), and valve disease or worsening heart failure hospitalizations (HR, 2.17; 95% CI, 1.23-3.84; $P = .008$) (**Figure 3**). Similar but less strong associations with clinical

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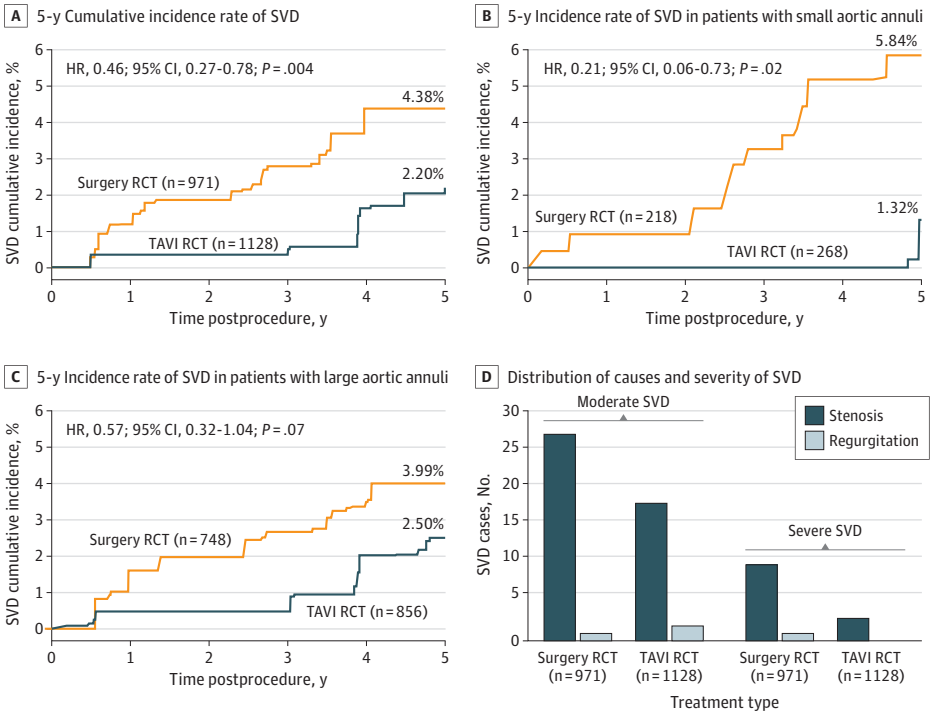
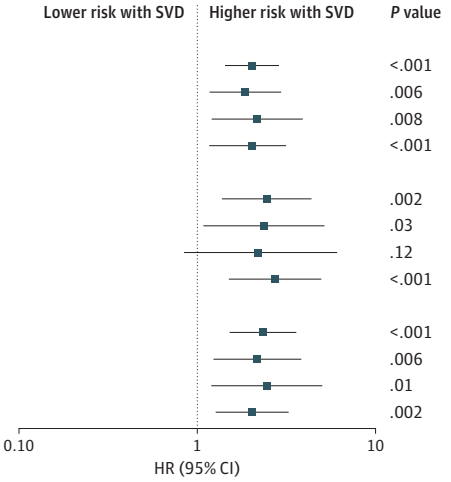
Figure 2. Comparison of Structural Valve Deterioration (SVD) in Patients Randomized to Surgery or Transcatheter Aortic Valve Implantation (TAVI)



Small aortic annuli was defined as computer tomography perimeter-derived diameter of 23 mm or smaller and large aortic annuli as greater than 23 mm. Severe SVD cases were based on status at any follow-up echocardiography, not just at last-available echocardiography. For hazard ratios (HRs), Fine-Gray *P* values are reported. AR indicates aortic regurgitation; AS, aortic stenosis; RCT, randomized clinical trial.

Figure 3. Association Between Clinical Outcomes and Structural Valve Deterioration (SVD)

Outcome	HR (95% CI)	*P* value
Pooled surgery RCT and all TAVI[a] (n = 4762)		
All-cause mortality	2.03 (1.46-2.82)	<.001
Cardiovascular mortality	1.86 (1.20-2.90)	.006
Hospitalization for AV disease/worsening HF	2.17 (1.23-3.84)	.008
Composite[b]	2.02 (1.42-2.88)	<.001
Surgery RCT (n = 971)		
All-cause mortality	2.45 (1.40-4.30)	.002
Cardiovascular mortality	2.37 (1.10-5.08)	.03
Hospitalization for AV disease/worsening HF	2.20 (0.81-5.98)	.12
Composite[b]	2.73 (1.53-4.88)	<.001
All TAVI[a] (n = 3791)		
All-cause mortality	2.34 (1.55-3.53)	<.001
Cardiovascular mortality	2.17 (1.26-3.76)	.006
Hospitalization for AV disease/worsening HF	2.45 (1.22-4.93)	.01
Composite[b]	2.03 (1.29-3.19)	.002



AV indicates aortic valve; HF, heart failure; HR, hazard ratio; RCT, randomized clinical trial; TAVI, transcatheter aortic valve implantation.

[a] The all TAVI cohort comprises the pooled RCT and non-RCT populations.

[b] Composite of all-cause mortality or hospitalization for AV disease or worsening HF.

outcomes were observed with other indices for SVD (eFigure 3 in the Supplement).

Predictors of SVD

Baseline clinical characteristics and univariate predictors of SVD are described in eTable 4 in the Supplement. Multivariate analysis found a higher risk of developing SVD in patients with a higher body surface area and a lower risk of SVD in men, older patients, and those with history of hypertension, percutaneous coronary intervention, and atrial fibrillation (**Figure 4**).

Discussion

Our pooled analysis of randomized patients found that the CoreValve/Evolut transcatheter bioprosthesis was associated with a lower rate of SVD compared with surgery at 5 years. This lower risk of SVD was most pronounced in patients with smaller aortic annuli (23 mm or smaller diameter). We also found that patients who developed SVD had a 2-fold higher 5-year mortality and hospitalizations for valve disease or worsening heart failure, suggesting that serial Doppler transthoracic echocar-

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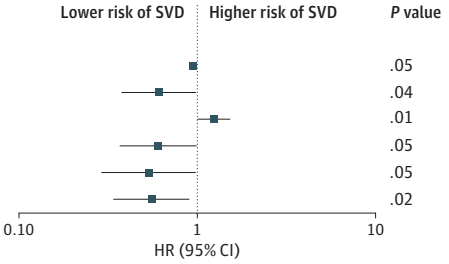
Figure 4. Multivariate Predictors of Structural Valve Deterioration (SVD)

Character	HR (95% CI)	P value
Pooled surgery RCT and all TAVI[a] (n = 4762)		
Age, y	0.97 (0.95-1.00)	.05
Male	0.62 (0.39-0.99)	.04
Body surface area, m^2[b]	1.28 (1.05-1.55)	.01
Prior percutaneous coronary intervention	0.62 (0.38-1.00)	.05
Hypertension	0.55 (0.30-0.99)	.05
Prior atrial fibrillation/flutter	0.57 (0.35-0.91)	.02



HR indicates hazard ratio; RCT, randomized clinical trial; TAVI, transcatheter aortic valve implantation.

[a] The all TAVI cohort comprises the pooled RCT and non-RCT populations.

[b] HR per 0.2-m^2 increase in body surface area.

diography is a valuable tool to monitor patients after aortic valve replacement, regardless of the modality of valve replacement or the definition used for SVD. Our multivariate analysis for preprocedural predictors of SVD identified that a larger body surface area was associated with higher rates of SVD, while men, older patients, and those with a history of hypertension, percutaneous coronary intervention, and atrial fibrillation had lower rates of SVD.

Competing Risk of Death and Bioprosthetic Valve Durability

Randomized studies comparing TAVI with surgery in patients with severe AS have shown that TAVI is an effective alternative to surgery in patients of all risk levels.[1-8] Patients enrolled in the Evolut Low Risk RCT had a median age of 74 years and 25% of patients were 70 years old or younger,[8] underscoring the importance of valve durability in the selection of the initial bioprosthetic valve in younger patients.[11]

Prior surgical series have shown that median life expectancy after surgical aortic valve replacement in low-risk patients varies by age; patients who are age 70 to 75 years at the time of surgery have an approximate 10-year to 13-year median life expectancy.[23] Johnston and colleagues[12] have demonstrated the competing risk of death influences the rate of surgical explant because of surgical valve failure and that surgical reintervention is required in a minority of patients during their lifetime.

An increase in valve durability may have an important influence on the need for subsequent reintervention in younger patients during their lifetime. Using death as a competing risk, this pooled analysis showed that SVD was lower in RCT patients undergoing TAVI than surgery (surgery, 4.38%; TAVI, 2.20%; HR, 0.46; 95% CI, 0.27-0.78; P = .004) at 5 years. This relative reduction in SVD was more profound in patients with a smaller annuli (less than 23 mm diameter; surgery, 5.84%; TAVI, 1.32%; HR, 0.21; 95% CI, 0.06-0.73; P = .02) but not in those with a larger annuli (larger than 23 mm diameter; surgery, 3.99%; TAVI, 2.50%; HR, 0.57; 95% CI, 0.32-1.04; P = .07). Accordingly, bioprosthetic valve durability, among other factors, should be an important consideration for the initial bioprosthetic valve choice in patients with severe AS.

SVD and Clinical Outcomes

Criteria have been proposed for defining bioprosthetic valve dysfunction after aortic valve replacement,[13,14] although none have been validated in clinical studies.[13,24,25] The Doppler-derived SVD definition used in this study, which is consistent with VARC-3 and European Association of Percutaneous Cardiovascular Interventions consensus documents, was associated with worsened 5-year clinical outcomes. A 2-fold increased all-cause mortality (HR, 2.03; 95% CI, 1.46-2.82; P < .001), cardiovascular mortality (HR, 1.86; 95% CI, 1.20-2.90; P = .006), and hospitalizations for aortic valve disease or worsening heart failure (HR, 2.17; 95% CI, 1.23-3.84; P = .008) was identified. These contemporary SVD criteria were more predictive of clinical outcomes than previously reported indices for HVD[20,21] or SVD that require more extensive hemodynamic criteria and documentation of associated smaller EOAs,[14] a factor that may be subject to substantial observer variability and error.[26]

Predictors of SVD

SVD after surgery has been shown to be more common in younger patients, women, in patients with a higher residual gradient, and in patients with end-stage kidney disease, among other factors.[12] Our analysis identified several important preprocedural predictors of SVD through 5 years. Patients who developed SVD were younger (79.4 years) than those who did not develop SVD (82.1 years; P = .003), similar to studies performed in patients treated with surgical valve replacement.[12,27] Women also developed SVD more often than men, and patients with prior PCI, atrial fibrillation and hypertension prior to aortic valve replacement had lower rates of SVD. This potentially may be because of antithrombotic or antihypertensive therapies; however, these postprocedural regimens were not systematically collected in the pooled studies and so we were not able to assess their effect on occurrence of SVD.

While some studies have suggested higher rates of surgical valve failure with different surgical bioprostheses,[28-30] the surgical valves used in this study reflect contemporary surgical practice. In more recent years, aortic root enlargement has allowed the use of a larger surgical bioprosthesis.[31] This trend should be considered in the interpretation of our results, although a reduction of SVD over time with annular enlargement has not been shown. Other RCTs have compared SVD in patients with AS undergoing TAVI and surgery.[15,32] The PARTNER II study found higher rates of 5-year SVD with the early-generation intra-annular, balloon-expandable valve than surgery, while a more contemporary balloon-expandable valve

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showed similar rates of SVD with TAVI vs surgery in nonrandomized patients.[32] In contrast, the all-comer NOTION RCT, a study of 280 patients with severe AS randomized to CoreValve bioprosthesis or surgery, found significantly lower rates of SVD after TAVI than surgery at 8 years (13.9% vs 28.3%; $P = .002$).[15]

Prior studies have also shown differences in SVD between balloon-expandable intra-annular bioprostheses and self-expanding supra-annular bioprostheses.[16,33,34] In the FRANCE-2 Registry, early-generation balloon-expandable intra-annular bioprostheses had higher rates of moderate SVD (13.8% vs 8.9% for CoreValve) and severe SVD (4.1% vs 0% for CoreValve).[33] In the randomized CHOICE trial, moderate or severe SVD occurred in 6.6% of early-generation balloon-expandable intra-annular transcatheter bioprostheses vs 0% in CoreValve bioprostheses ($P = .02$).[34] A meta-analysis of published studies demonstrated self-expanding TAVI had the lowest risk of SVD compared with balloon-expandable TAVI and surgery at mid-term follow-up.[16] Comparative RCTs are needed, and an ongoing study will randomize 700 patients with a small annulus area less than 430 mm^2 to TAVI with the intra-annular SAPIEN 3/3 Ultra bioprostheses (Edwards Lifesciences) or the supra-annular Evolut PRO/PRO+ bioprostheses.[35]

Limitations

There are several limitations to the current study. This post hoc analysis was intended to focus on SVD. Although nonstructural valve dysfunction and bioprosthetic valve failure are definitions that provide a more complete picture of bioprosthetic valve durability, they are out of the scope of this study. Such investigations are planned for future work. Morphological valve deterioration data were not systematically collected in the pooled studies, and 2-dimensional transthoracic echocardiograms provided limited visualization of the morphological aspects of the leaflets; therefore, this study evaluated moderate or greater SVD related to hemodynamic deterioration. Our primary SVD definition used objective hemodynamic criteria for the development of AS or regurgitation based on VARC-3[14] but did not include concurrent changes in EOA or Doppler velocity index. We cannot exclude a possibility that SVD was related to changes in stroke volume in some patients, but this factor did not influence the overall conclusions of this study. Moreover, the contemporary SVD definition used in this pooled analysis provided a more robust prediction of clinical outcomes compared with the complete VARC-3 SVD definition.[14] The lack of complete serial echocardiographic follow-up examinations from the Core Laboratory was a limitation of this study; 57 of 95 patients (RCT, 16 receiving surgery and 6 receiving TAVI; non-RCT, 35 receiving TAVI) who developed SVD were identified by site-reported echocardiographic readings. However, an algorithm established by a group of 5 experts that evaluated all Core Laboratory and site-reported echocardiographic parameters verified all potential SVD cases. This post hoc analysis included older patients, with a mean (SD) age of 82.1 (7.4) years; therefore, further analysis in younger patients is warranted. The competing risk of mortality limited the number of participants with SVD, similar to prior surgical trials. Current follow-up is limited to 5 years, and 10-year follow-up is ongoing for the SURTAVI and Low Risk RCTs. Additionally, it should be noted that most of the patients receiving TAVI were implanted with the early-generation CoreValve bioprosthesis, as only 11.5% of the RCT patients received the Evolut R bioprosthesis.

Conclusions

We found that the cumulative incidence of SVD was significantly lower among randomized patients treated with a self-expanding supra-annular transcatheter valve than surgery. Doppler-derived SVD was associated with a 2-fold increased risk of late mortality and hospitalizations for valve disease or worsening heart failure. Although long-term 10-year follow-up is ongoing, valve durability using clinically relevant SVD criteria should be an important consideration for the selection of the first bioprosthetic valve in lower-risk patients with symptomatic severe AS.

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